TECHNO INDIA COLLEGE OF TECHNOLOGY

Department of Computer Science & Engineering

Computer Network Lab Manual

Paper Code: CS692

Objective: Get acquainted with some commonly used networking commands, network services and TCP/IP diagnostic tools		
LAB 3	1. Familiarization to Commands like *ifconfig, traceroute, arp, ping, netstat, route add, tcpdump* etc. (Linux) 2. Installing the NIC Card. Network trouble shooting Techniques. 3. Familiarization to various network services and applications like ftp, telnet, ssh, dns, dhcp	3 Hours

Familiarization to network Commands in Linux

To display all system interfaces, use the *ifconfig -a* command:

```
#ifconfig -a
```

When this command is run on Linux, the following sample is shown

```
eth0      Link encap:Ethernet  HWaddr 00:10:5A:28:5D:7C
          inet addr:10.0.2.201  Bcast:10.0.2.255  Mask:255.255.255.0
UP BROADCAST RUNNING PROMISC MULTICAST  MTU:1500  Metric:1
RX packets:822 errors:1 dropped:0 overruns:0 frame:1
     TX packets:108 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:100
          Interrupt:10 Base address:0xfc00          Physical hardware interface

lo        Link encap:Local Loopback                 Loopback interface for
          inet addr:127.0.0.1  Mask:255.0.0.0        internal diagnostics
          UP LOOPBACK RUNNING  MTU:3924  Metric:1
          RX packets:92 errors:0 dropped:0 overruns:0 frame:0
          TX packets:92 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:0                 Dial-up interface for
                                                    ISP connectivity
ppp0      Link encap:Point-to-Point Protocol
          POINTOPOINT NOARP MULTICAST  MTU:1500  Metric:1
          RX packets:0 errors:0 dropped:0 overruns:0 frame:0
          TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:10
```

To place an interface in the down state, invoke the *ifconfig* command with the appropriate interface and the down keyword option as shown

 # ifconfig eth0 down

To enable or activate this interface, we simply use the up command option:

 # ifconfig eth0 up

To find the hostname of your system

 # hostname

To set a new hostname

 # *hostname* new_hostname

To change the interface configuration, the administrator could use From Shell

 # ifconfig eth0 128.197.10.1 netmask 255.255.0.0 broadcast 128.197.255.255

Show / manipulate the IP routing table on Linux using `route` command

 # route **or**

 # route –n **or**

 # netstat –r **or**

 # ip route show **or**

 # ip route list

Sample outputs:

```
root@backup2:~# route
Kernel IP routing table
Destination     Gateway         Genmask         Flags Metric Ref    Use Iface
default         5.153.23.97-sta 0.0.0.0         UG    100    0        0 eth1
5.153.23.96     *               255.255.255.248 U     0      0        0 eth1
10.0.0.0        10.70.203.65    255.0.0.0       UG    0      0        0 eth0
10.70.203.64    *               255.255.255.192 U     0      0        0 eth0
root@backup2:~# route -n
Kernel IP routing table
Destination     Gateway         Genmask         Flags Metric Ref    Use Iface
0.0.0.0         5.153.23.97     0.0.0.0         UG    100    0        0 eth1
5.153.23.96     0.0.0.0         255.255.255.248 U     0      0        0 eth1
10.0.0.0        10.70.203.65    255.0.0.0       UG    0      0        0 eth0
10.70.203.64    0.0.0.0         255.255.255.192 U     0      0        0 eth0
root@backup2:~#
```

Fig.01: Display routing table using route command

Linux add a default route using `route` command

Route all traffic via 192.168.1.254 gateway connected via eth0 network interface:

 # route add default gw 192.168.1.254 eth0

Verify newly added route ip in the Linux kernel routing table

To verify new routing table, enter:

 # ip route list

Displaying Active Network Sessions: To display the currently established connections, issue the *netstat* command with the –t option as shown here:

```
# netstat -t
Active Internet connections (w/o servers)
Proto Recv-Q Send-Q Local Address           Foreign Address         State
tcp      0    0 110.orlando-11-12r:1052 192.215.123.37:www      ESTABLISHED
tcp      1    0 110.orlando-11-12r:1051 192.215.123.37:www      CLOSE
tcp      0    6 110.orlando-11-12r:1050 postoffice.worldn:pop-3 ESTABLISHED
tcp      0    0 110.orlando-11-12r:1049 www3.yahoo.com:www      ESTABLISHED
tcp      0    0 socrates.home.co:telnet durer.home.com:1033     ESTABLISHED
tcp      0    0 socrates.home.co:telnet durer.home.com:1032     ESTABLISHED
```

Displaying Interface Information: The *netstat* command can obtain details on the configuration of the network Interface. The *-i* command-line option obtains a list of each defined interface on the system

```
#netstat -I
Kernel Interface table
Iface   MTU Met    RX-OK RX-ERR RX-DRP RX-OVR    TX-OK TX-ERR TX-DRP TX-OVR Flg
eth0   1500   0        0      0      0      0        0      1      0      0 BRU
eth0:  1500   0      - no statistics available -                            BRU
eth1   1500   0     3946      0      0      0      138      0      0      0 BRU
lo     3924   0      192      0      0      0      192      0      0      0 LRU
```

Display Protocol Statistics: The *netstat* command can be used to display protocol statistics. The statistics option, by itself, will display the supported protocols, including TCP, UDP, and RAW. RAW is a combination of both IP and ICMP packets and can be displayed separately using the keyword raw.

```
# netstat -s
Ip:
    3003 total packets received
    0 forwarded
    0 incoming packets discarded
    212 incoming packets delivered
    2847 requests sent out
Icmp:
    489 ICMP messages received
    0 input ICMP message failed.
    ICMP input histogram:
        destination unreachable: 486
        echo replies: 3
    487 ICMP messages sent
    0 ICMP messages failed
    ICMP output histogram:
        destination unreachable: 487
Tcp:
```

Verify Network Connectivity Using Ping: The ping tool can be used to determine general availability of any TCP/IP device, even if it doesn't specifically have a general operating system. For example, to determine if the host *durer* is reachable, issue the following ping command:

```
#ping durer
PING durer.home.com (10.0.2.10): 56 data bytes
64 bytes from 10.0.2.10: icmp_seq=0 ttl=128 time=0.9 ms
64 bytes from 10.0.2.10: icmp_seq=1 ttl=128 time=0.8 ms
64 bytes from 10.0.2.10: icmp_seq=2 ttl=128 time=0.8 ms
64 bytes from 10.0.2.10: icmp_seq=3 ttl=128 time=0.8 ms

--- durer.home.com ping statistics ---
4 packets transmitted, 4 packets received, 0% packet loss
round-trip min/avg/max = 0.8/0.8/0.9 ms
```

Show Basic Network Performance: The ping command can be used to measure the amount of time required to transmit a message to a remote destination and the time required to obtain a response. Here –s option issue a single ICMP request per second

Target IP address

Round trip time

```
# ping -s didymus
PING didymus.home.com (10.0.2.127): 56 data bytes
64 bytes from 10.0.2.127: icmp_seq=0 ttl=255 time=1.2 ms
64 bytes from 10.0.2.127: icmp_seq=1 ttl=255 time=1.1 ms
64 bytes from 10.0.2.127: icmp_seq=2 ttl=255 time=1.2 ms
64 bytes from 10.0.2.127: icmp_seq=3 ttl=255 time=1.2 ms

--- didymus.home.com ping statistics ---
4 packets transmitted, 4 packets received, 0% packet loss
round-trip min/avg/max = 1.1/1.1/1.2 ms
```

4 packets sent and received

ICMP sequence number

Packet size

```
# ping -s 100 didymus
```

The command issues the ICMP requests with a packet size of 100 bytes to the target host *didymus*.

Show Network Connectivity using *traceroute* :

The *traceroute* command examines and records the path to a specified network destination.

To display the IP path between the local system running *traceroute* and the destination system called *vermeer*, issue the following command:

 # traceroute vermeer

```
# traceroute Vermeer
traceroute to vermeer (128.197.2.200), 30 hops max, 40 byte packets
 1  Router-Z (10.0.2.129)  4.256 ms *  2.899 ms
 2  vermeer (128.197.2.200)  7.340 ms  7.433 ms  7.526 ms
```

Measuring Network Performance using *traceroute:* Given the fact that at any point in time the performance or RTT between networks and systems can change significantly

```
# traceroute 198.137.240.91
traceroute to 198.137.240.91 (198.137.240.91), 30 hops max, 40 byte packets
 1  10.0.2.76 (10.0.2.76)  19.906 ms  9.801 ms  8.174 ms
 2  199.70.195.38 (199.70.195.38)  197.460 ms  188.000 ms  181.029 ms
 3  12.77.194.1 (12.77.194.1)  166.802 ms  184.713 ms  185.857 ms
 4  12.127.12.205 (12.127.12.205)  245.026 ms  270.253 ms  266.718 ms
 5  12.127.15.145 (12.127.15.145)  215.191 ms  211.920 ms  208.979 ms
 6  192.205.31.165 (192.205.31.165)  217.875 ms  232.610 ms  222.274 ms
 7  204.6.117.65 (204.6.117.65)  266.797 ms  239.000 ms  215.671 ms
 8  38.1.4.69 (38.1.4.69)  235.431 ms  225.447 ms  301.119 ms
 9  38.1.25.5 (38.1.25.5)  235.369 ms  236.134 ms  263.557 ms
10  38.1.25.5 (38.1.25.5)  252.172 ms  238.984 ms  263.013 ms
11  38.146.148.45 (38.146.148.45)  241.956 ms  248.091 ms  243.300 ms
12  198.137.240.33 (198.137.240.33)  249.361 ms  228.717 ms  252.927 ms
13  198.137.240.91 (198.137.240.91)  238.799 ms  259.967 ms  236.384 ms
```

When the trace is repeated later, it shows the following:

```
traceroute to 198.137.240.91 (198.137.240.91), 30 hops max, 40 byte packets
 1  10.0.2.76 (10.0.2.76)  7.619 ms  5.863 ms  6.206 ms
 2  199.70.195.42 (199.70.195.42)  177.685 ms  177.691 ms  177.842 ms
 3  12.77.242.129 (12.77.242.129)  170.712 ms  177.096 ms  173.517 ms
 4  12.127.12.205 (12.127.12.205)  260.239 ms  248.072 ms  252.829 ms
 5  12.127.15.145 (12.127.15.145)  219.767 ms  215.645 ms  232.399 ms
 6  192.205.31.165 (192.205.31.165)  232.259 ms  225.243 ms  219.236 ms
 7  204.6.117.65 (204.6.117.65)  228.997 ms  218.067 ms  219.365 ms
 8  38.1.4.69 (38.1.4.69)  445.758 ms  232.797 ms  276.249 ms
 9  38.1.25.5 (38.1.25.5)  245.674 ms  443.611 ms  577.309 ms
10  38.1.25.5 (38.1.25.5)  432.994 ms  222.527 ms  242.844 ms
11  38.146.148.45 (38.146.148.45)  257.668 ms  249.923 ms  263.074 ms
12  198.137.240.33 (198.137.240.33)  276.658 ms  242.361 ms *
13  198.137.240.91 (198.137.240.91)  248.266 ms  245.006 ms  251.071 ms
```

Significant RTT delays

Monitor Network Traffic with the *tcpdump:*

The *tcpdump* command is a general-purpose network traffic monitor that can capture and display packets and their contents. This command can be used as a protocol analyzer, providing one of the best ways to investigate connectivity problems among systems and networking devices.

When *tcpdump* is invoked without command-line options, it opens the primary network interface and begins capturing frames from the local network and displaying their contents.

 # tcpdump or

 # tcpdump –i eth0 –w filename

Examples:

To capture all packets from the IP address 10.0.2.100, use

 # tcpdump ipaddr 10.0.2.100

To display all packets that are greater than 56 bytes, invoke the following command:

 # tcpdump greater 56

To display all packets that are less than 60 bytes, use the less primitive:

 # tcpdump -x less 60

To display all the traffic involving monet, which includes packets being sent to and received from monet

 # tcpdump host monet

To display traffic received from monet

 # tcpdump from host monet

To display all Address Resolution Protocol packets. This includes both requests and replies.

 # tcpdump arp

To display all Transmission Control Protocol (TCP) packets from/to the host durer.

#tcpdump host durer and tcp

To display all packets using port 23 from or to host vectra. This amounts to inspecting all Telnet packets going to this system from others on the network. Recall that port 23 is the Telnet service port for all incoming packets.

#tcpdump host vectra and port 23

Displaying the ARP Cache: To display the contents of the ARP table, use the arp -v command. When the table is displayed, it includes the device name, hardware type, physical address, mask flag, and network interface.

```
# arp -a
Address              Hwtype    HWaddress          Flags Mask   Iface
durer.home.com        ether     00:60:97:0E:A3:06     C          eth0
switch.home.com       ether     00:00:1D:0A:5D:01     C          eth0
probe.home.com        ether     00:C0:D4:00:09:40     C          eth0
rembrandt.home.com    ether     08:00:20:82:be:05     C          eth0
rubens.home.com                  (incomplete)                    eth0
monet.home.com        ether     08:00:20:04:cf:2c     C          eth0
Entries: 6 Skipped: 0          Found: 6
```

Option	Description
-a	Displays the current ARP entry for a specific host (Linux only). Displays all of the entries within the ARP table (Solaris and HP-UX).
-d	Deletes an ARP entry specified after this option.
-f	Loads a file that contains entries to place in the cache.
-i	Displays only those entries for specified interface (Linux only).
-n	Shows numerical addresses instead of hostnames (Linux only).
-s	Creates an ARP entry.
-v	Displays ARP cache using verbose mode (Linux only).

Examples:

The following Linux command will list the ARP mapping for the host called ***socrates***:

```
# arp -a Socrates
socrates.home.com (10.0.2.201) at 00:60:97:0E:A3:03 [ether] on eth0
```

To display all ARP listings for the eth1 interface:

```
#arp -i eth1
```

To remove the host ***durer*** is from the ARP table

```
# arp -d durer
```

To add an ARP entry, use the -s option followed by the hostname (or address) and the associated physical data link address. For example, let's say we would like to add a system called ***bruegel*** to the ARP table

```
# arp -s bruegel 08:00:20:82:be:05
```

Nmap :

Nmap is a powerful and versatile security scanner. Although we rely on it for many things, we use it mainly as a port scanner, to see which ports that are open on a particular host. Please use nmap with discretion if you use it to scan computers and networks other than your own.

nmap 192.168.10.1

Install in Ubuntu: `sudo apt-get install nmap`

nslookup:

The nslookup command provides a mechanism to query a DNS server from the command line. The command has two basic modes: simple query and complex command interface mode. **To use simple query mode**, provide a hostname for DNS resolution as an option to the command (Linux):

```
# nslookup dino.home.com
Note:  nslookup is deprecated and may be removed from future releases.
Consider using the `dig' or `host' programs instead.  Run nslookup with
the `-sil[ent]' option to prevent this message from appearing.
Server:         216.155.202.163
Address:        216.155.202.163#53

Name:   dino.home.com
Address: 216.155.202.110
```

To use complete query mode; issue the nslookup command without any options:

```
# nslookup
Default Server:  didymus.home.com
Address:  216.155.202.163

>
```

host:

The host command is relatively new to the DNS community and comes with the basic Linux operating system. It supports basic hostname resolution. Thus, using this command with a valid DNS name yields

```
# host dino
dino.home.com. has address 216.155.202.110
```

Network Configuration in Ubuntu Server version 14.04

Ubuntu ships with a number of graphical utilities to configure your network devices. This document is geared toward server administrators and will focus on managing your network on the command line.

Ethernet Interfaces

Ethernet interfaces are identified by the system using the naming convention of ethX, where X represents a numeric value. The first Ethernet interface is typically identified as eth0, the second as eth1, and all others should move up in numerical order.

Identify Ethernet Interfaces

To quickly identify all available Ethernet interfaces, you can use the ifconfig command as shown below.

```
ifconfig -a | grep eth

eth0      Link encap:Ethernet  HWaddr 00:15:c5:4a:16:5a
```

IP Addressing

The following section describes the process of configuring your systems IP address and default gateway needed for communicating on a local area network and the Internet.

Temporary IP Address Assignment

For temporary network configurations, you can use standard commands such as ip, ifconfig and route, which are also found on most other GNU/Linux operating systems. These commands allow you to configure settings which take effect immediately, however they are not persistent and will be lost after a reboot.

To temporarily configure an IP address, you can use the ifconfig command in the following manner. Just modify the IP address and subnet mask to match your network requirements.

```
sudo ifconfig eth0 10.0.0.100 netmask 255.255.255.0
```

To verify the IP address configuration of eth0, you can use the ifconfig command in the following manner.

```
ifconfig eth0
eth0      Link encap:Ethernet  HWaddr 00:15:c5:4a:16:5a
```

```
inet addr:10.0.0.100  Bcast:10.0.0.255  Mask:255.255.255.0
inet6 addr: fe80::215:c5ff:fe4a:165a/64 Scope:Link
UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
RX packets:466475604 errors:0 dropped:0 overruns:0 frame:0
TX packets:403172654 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:2574778386 (2.5 GB)  TX bytes:1618367329 (1.6 GB)
Interrupt:16
```

To configure a default gateway, you can use the route command in the following manner. Modify the default gateway address to match your network requirements.

```
sudo route add default gw 10.0.0.1 eth0
```

To verify your default gateway configuration, you can use the route command in the following manner.

```
route -n

Kernel IP routing table
Destination     Gateway         Genmask         Flags Metric Ref    Use Iface
10.0.0.0        0.0.0.0         255.255.255.0   U     1      0        0 eth0
0.0.0.0         10.0.0.1        0.0.0.0         UG    0      0        0 eth0
```

If you no longer need this configuration and wish to purge all IP configuration from an interface, you can use the ip command with the flush option as shown below.

```
ip addr flush eth0
```

Dynamic IP Address Assignment (DHCP Client)

To configure your server to use DHCP for dynamic address assignment, add the dhcp method to the inet address family statement for the appropriate interface in the file /etc/network/interfaces. The example below assumes you are configuring your first Ethernet interface identified as eth0.

```
auto eth0
iface eth0 inet dhcp
```

By adding an interface configuration as shown above, you can manually enable the interface through the ifup command which initiates the DHCP process via dhclient.

```
sudo ifup eth0
```

To manually disable the interface, you can use the ifdown command, which in turn will initiate the DHCP release process and shut down the interface.

```
sudo ifdown eth0
```

Static IP Address Assignment

To configure your system to use a static IP address assignment, add the static method to the inet address family statement for the appropriate interface in the file /etc/network/interfaces. The example below assumes you are configuring your first Ethernet interface identified as eth0. Change the address, netmask, and gateway values to meet the requirements of your network.

```
auto eth0
iface eth0 inet static
address 10.0.0.100
netmask 255.255.255.0
gateway 10.0.0.1
```

By adding an interface configuration as shown above, you can manually enable the interface through the ifup command.

```
sudo ifup eth0
```

To manually disable the interface, you can use the ifdown command.

```
sudo ifdown eth0
```

Loopback Interface

The loopback interface is identified by the system as lo and has a default IP address of 127.0.0.1. It can be viewed using the ifconfig command.

```
ifconfig lo
lo        Link encap:Local Loopback
          inet addr:127.0.0.1  Mask:255.0.0.0
          inet6 addr: ::1/128 Scope:Host
          UP LOOPBACK RUNNING  MTU:16436  Metric:1
          RX packets:2718 errors:0 dropped:0 overruns:0 frame:0
          TX packets:2718 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:0
          RX bytes:183308 (183.3 KB)  TX bytes:183308 (183.3 KB)
```

By default, there should be two lines in /etc/network/interfaces responsible for automatically configuring your loopback interface. It is recommended that you keep the default settings unless you have a specific purpose for changing them. An example of the two default lines are shown below.

```
auto lo
iface lo inet loopback
```

Permanent IP Address Assignment

/etc/network/interfaces file contains network interface configuration information for the both Ubuntu and Debian Linux. This is where you configure how your system is connected to the network.

Your main network configuration file is */etc/network/interfaces*

Desired new sample settings:
=> Host IP address 192.168.1.100
=> Netmask: 255.255.255.0
=> Network ID: 192.168.1.0
=> Broadcast IP: 192.168.1.255
=> Gateway/Router IP: 192.168.1.254
=> DNS Server: 192.168.1.254

Open network configuration file

$ **sudo vi /etc/network/interfaces** OR

$ **sudo nano /etc/network/interfaces**

Append new network settings:

```
iface eth0 inet static
address 192.168.1.100
netmask 255.255.255.0
network 192.168.1.0
broadcast 192.168.1.255
gateway 192.168.1.254
dns-nameservers 192.168.1.254
```

Save and close the file. Restart the network:

```
$ sudo /etc/init.d/networking restart
```

Name Resolution

Name resolution as it relates to IP networking is the process of mapping IP addresses to hostnames, making it easier to identify resources on a network. The following section will explain how to properly configure your system for name resolution using DNS and static hostname records.

DNS Client Configuration

Traditionally, the file **/etc/resolv.conf** was a static configuration file that rarely needed to be changed or automatically changed via DCHP client hooks. Nowadays, a computer can switch from one network to another quite often and the resolvconf framework is now being used to track these changes and update the resolver's configuration automatically. It acts as an intermediary between programs that supply nameserver information and applications that need nameserver information. Resolvconf gets populated with information by a set of hook scripts related to network interface configuration. The most notable difference for the user is that any change manually done to /etc/resolv.conf will be lost as it gets overwritten each time something triggers resolvconf. Instead, resolvconf uses DHCP client hooks, and /etc/network/interfaces to generate a list of nameservers and domains to put in /etc/resolv.conf, which is now a symlink:

```
/etc/resolv.conf -> ../run/resolvconf/resolv.conf
```

To configure the resolver, add the IP addresses of the nameservers that are appropriate for your network in the file /etc/network/interfaces. You can also add an optional DNS suffix search-lists to match your network domain names. For each other valid resolv.conf configuration option, you can include, in the stanza, one line beginning with that option name with a dns- prefix. The resulting file might look like the following:

```
iface eth0 inet static
address 192.168.3.3
netmask 255.255.255.0
gateway 192.168.3.1
dns-search example.com
dns-nameservers 192.168.3.45 192.168.8.10
```

The search option can also be used with multiple domain names so that DNS queries will be appended in the order in which they are entered. For example, your network may have multiple sub-domains to search; a parent domain of example.com, and two sub-domains, sales.example.com and dev.example.com.

If you have multiple domains you wish to search, your configuration might look like the following:

```
iface eth0 inet static
address 192.168.3.3
netmask 255.255.255.0
gateway 192.168.3.1
dns-search example.com sales.example.com dev.example.com
dns-nameservers 192.168.3.45 192.168.8.10
```

If you try to ping a host with the name of server1, your system will automatically query DNS for its Fully Qualified Domain Name (FQDN) in the following order:

1. server1.example.com
2. server1.sales.example.com
3. server1.dev.example.com

If no matches are found, the DNS server will provide a result of notfound and the DNS query will fail.

Static Hostnames

Static hostnames are locally defined hostname-to-IP mappings located in the file **/etc/hosts**. Entries in the hosts file will have precedence over DNS by default. This means that if your system tries to resolve a hostname and it matches an entry in **/etc/hosts**, it will not attempt to look up the record in DNS. In some configurations, especially when Internet access is not required, servers that communicate with a limited number of resources can be conveniently set to use static hostnames instead of DNS.

The following is an example of a hosts file where a number of local servers have been identified by simple hostnames, aliases and their equivalent Fully Qualified Domain Names (FQDN's).

```
127.0.0.1       localhost
127.0.1.1       ubuntu-server
10.0.0.11       server1 server1.example.com vpn
10.0.0.12       server2 server2.example.com mail
10.0.0.13       server3 server3.example.com www
10.0.0.14       server4 server4.example.com file
```

In the above example, notice that each of the servers have been given aliases in addition to their proper names and FQDN's. Server1 has been mapped to the name vpn, server2 is referred to as mail, server3 as www, and server4 as file.